A CLIPS Rule-Based Agent for Wumpus World

Planning, inference and bounded risk in a dark cave

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The problem

- A grid cave, unknown layout, local percepts only.
- **breeze**: an adjacent pit. **stench**: an adjacent Wumpus. **glitter**: the gold is here.
- Every action costs energy; bumps cost extra. `quit` is valid only at `(0, 0)`.
- The Wumpus is unique; the gold is unique; a pit may share a cell with a wall.

Goals, in priority order: survive → collect the gold → return to `(0, 0)` and quit → preserve energy → test pit hypotheses.

Simulator and protocol

Action	Effect	Energy	Reward
`up/down/left/right`	move if free, otherwise bump and stay	−1 / −2	step cost
`grab`	collect the gold in the current cell	−1	+100
`quit`	end the episode	−1	0
`pit-*`	test the adjacent cell, no movement	−1	+25 / −25

- Exactly one `(action (move ?))` per cycle; old percepts are retracted.
- The initial percept is asserted before the first `run()`.
- A quit outside `(0, 0)` is reported as a protocol error.

Architecture: six phases, one cycle

```
MAIN  (schema, counter, stale cleanup, dispatch)
  |
  +--> PERCEPT   percept -> pos + visited
  +--> MAP       visited/blocked -> no-pit, no-wumpus, safe, nbr, frontier
  +--> HYPO      breeze/stench closure, candidates, pit/Wumpus, CF
  +--> DECIDE    home/dist/nav, energy budget, goal, next hop
  +--> ACT       exactly one action: grab, quit, guess, move
```

- `MAIN` focuses `PERCEPT MAP HYPO DECIDE ACT`; the focus stack is one-way.
- Every action rule carries `(not (action (move ?)))`.
- No `halt`: the agenda drains naturally.

The fact model

Fact	Meaning
`percept`, `counter`, `action`	bridge input/output and clock
`pos`, `visited`, `blocked`	current cell, explored cells, walls and out-of-map cells
`no-pit`, `no-wumpus`, `safe`, `frontier`	classical exclusions, hazard-free cells, safe-unvisited leaves
`candidate`, `pit-hypothesis`	a breeze explanation and its CF evidence
`pit`, `wumpus`, `wumpus-possible`	confirmed hazards and the Wumpus intersection
`home`, `dist`, `nav`, `goal`, `next`	return distance, current distance, goal distance, target, ranked hop
`guessed`, `has-gold`, `probes-used`	persistent mission state

Hazard inference

- Visited → not a pit, not the Wumpus, safe.
- No breeze → every orthogonal neighbour is `no-pit` (walls included).
- No stench → every orthogonal neighbour is `no-wumpus`.
- A bump blocks a cell and rules out the Wumpus, but **not** a pit: a pit may hide under a wall.
- Breeze → candidates = adjacent cells that are not `no-pit` and not the Wumpus.
- One candidate for a breeze → confirmed pit. Confirmed pits stay candidates.
- Wumpus candidates must be adjacent to **every** stench cell; one survivor → confirmed Wumpus.

The Wumpus intersection uses a negated conjunction: “no stench cell lacks this neighbour”. This avoids a derive/prune loop.

Certainty factors with explicit slots

- CLIPS 6.41 has no native CFs: `(declare (CF ...))` and `(get-cf ...)` fail.
- Each breeze contributes `0.4` to every surviving candidate, once.
- `source-x`/`source-y` keep evidence facts distinct under CLIPS fact deduplication.
- Same target → MYCIN positive combination:

$$\mathrm{cf} = \mathrm{cf}_1 + \mathrm{cf}_2(1 - \mathrm{cf}_1)$$

- `0.4` → `0.64` → `0.784`; a confirmed pit is raised to `1.0`.
- CF is only a ranking/annotation: it never creates `safe` or `pit`.

Planning and energy

- `home`: shortest return distance over **visited** cells; exact and bump-proof.
- `dist`: shortest distance from the current cell; `nav`: distance to the chosen goal.
- Expand/relax to a fixed point; unit costs, deterministic tie-break.
- Affordability:

$$\mathrm{dist}(f) + \mathrm{home}(f) + 1 + \mathrm{reserve} \leq \mathrm{energy}$$

  , reserve = 5.
- Goal priority: grab → probe → quit → exit → frontier.
- Grab only if the grab plus the return still leaves the reserve.
- Next hop: a neighbour with `nav = nav(pos) - 1`; rank `up=1`, `right=2`, `down=3`, `left=4`.
- A bump is detected by comparing the intended move with the next position.

map1: a dead end that needs a bounded risk

```
safe closure from (0,0):

  (0,0) (0,1) (0,2) (0,3) (0,4)
              (1,3) (1,4) (2,4)

  the only connection is (3,4)
  (2,4) breeze: (2,3) pit  (2,5) pit  (3,4) safe
  -> no sound rule can prove (3,4) safe
```

- Last resort only: no safe frontier, no gold yet, return reserve intact.
- At most two probes; the second unlocks only after the first reached its cell.
- Candidate ceiling `cf <= 0.75`; exclude blocked, Wumpus-possible, guessed.
- Rank by visited-neighbour support, then CF, then geometry.
- `map1`: probe `(3,4)`, then `(4,3)`; both safe. `map10`: the probe hits a real pit — the accepted risk.

Pit guesses

- `pit-*` costs one energy; `+25` if the adjacent cell is a pit, `-25` otherwise.
- $$\mathrm{EV} = 25p - 25(1-p) - 1 = 50p - 26$$

  ; positive for $p > 0.52$.
- Conservative threshold: `cf >= 0.75`.
- Adjacent only, once per cell, never on blocked/safe/Wumpus-possible targets.
- Never while holding the gold, returning, or probing: mission goals have priority.
- Energy guard: `energy > home + reserve + 1`.
- Directions: `up -> pit-north`, `right -> pit-east`, `down -> pit-south`, `left -> pit-west`.

Evaluation: expert vs random

Agent	Maps	Survival	Gold	Return	Avg E	Guesses	Reward
expert	11	90.9%	81.8%	90.9%	29.8	7/7	975
random	11	0.0%	0.0%	0.0%	−0.1	0/0	−1100

- `map1`: 76 steps, reward 125, energy 111, gold, return, one correct guess `(2,3)`.
- `map7`: gold unaffordable under the 14-step budget; the agent returns and quits.
- `map10`: the bounded probe hits a real pit; the alternative safe crossing is not chosen.
- No protocol errors on the suite.

Limitations and conclusion

Limitations

- The risk probe can die by design; it is a last resort, not a proof.
- CFs are heuristic; overlapping breezes can inflate a safe cell.
- Pit guesses are adjacent-only and can be wrong.
- The return field is conservative (visited graph only).

Conclusion

- A complete, explainable CLIPS agent: six modules, classical inference, explicit MYCIN CFs, distance-field planning, bounded risk, pit guesses.
- The simulator and harness are headless, deterministic and tested.
- On 11 maps: 10 survivals, 9 golds, 10 returns, 975 points; the random baseline scores −1100.

Thank you — questions?